

ENTERPRISE SPEC

IEEE 830-1998 ALIGNED

REVISION 3.2

Software Requirements Specification

Hyperlocal Food Delivery & Restaurant Aggregation Platform (Zomato Ecosystem)

Document Standard	Target Architecture	Core Protocols	Status / Confidentiality
IEEE Std 830-1998 / 29148	Event-Driven Cloud Microservices	REST, WSS, PostGIS, TLS 1.3	Finalized / Technical Baseline

1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) establishes the functional, behavioral, performance, and infrastructural baseline for the Zomato Hyperlocal Food Delivery and Restaurant Aggregation Platform. It forms a formal contract across engineering squads, system architects, QA verification teams, product operations, and platform executives. This document models the complete lifecycle of multi-party order brokering across customer endpoints, merchant dashboards, delivery partner nodes, and central administrative clearing houses.

1.2 Scope

The system is a distributed, location-aware digital commerce and logistics orchestration engine.

OPERATIONAL BOUNDARIES

- System Boundaries (In-Scope):** Real-time merchant discovery via spatial queries; cart concurrency protection; multi-tier tax, surge, and distance pricing algorithms; payment gateway offloading; bi-directional WebSocket telemetry streaming; automated courier dispatch heuristics; merchant inventory toggling; and verified review workflows.
- Out-of-Scope Constraints:** Physical kitchen culinary operations, proprietary Kitchen Display System (KDS) physical fabrication, local vehicle maintenance, and direct on-premise PCI-DSS primary cardholder storage.

1.3 Definitions, Acronyms, and Abbreviations

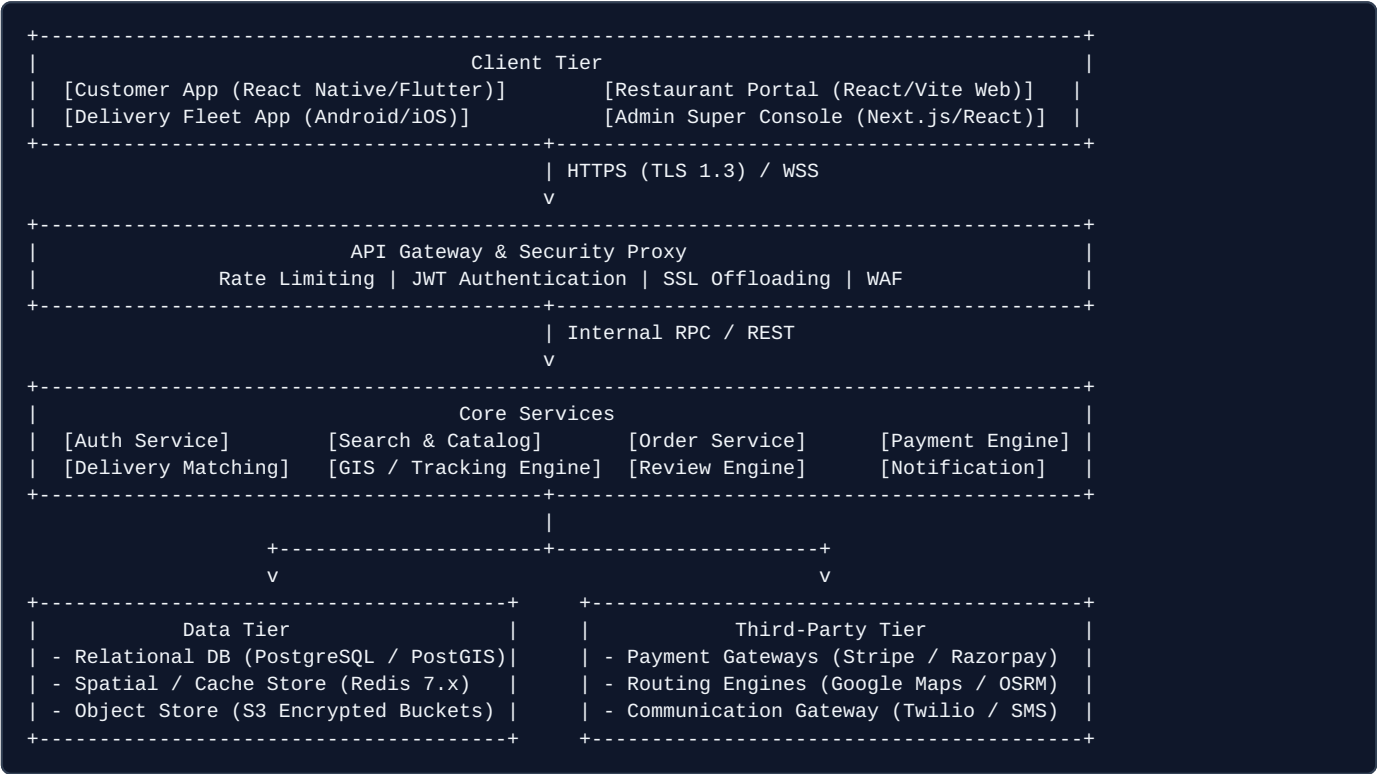
Acronym / Term	Engineering Definition
API / REST	Application Programming Interface complying with Representational State Transfer constraints over HTTPS.
ETA	Estimated Time of Arrival derived dynamically from routing networks, live traffic telemetry, and preparation buffers.
GIS / PostGIS	Geographic Information Systems; spatial database extension for PostgreSQL supporting geometry and geography calculations.
JWT	JSON Web Token (RFC 7519) utilizing cryptographic signatures (HMAC-SHA256 or RSA-256) for stateless authentication.
OTP	One-Time Password; ephemeral cryptographically random numeric string used for multi-factor login and drop verification.
PCI-DSS	Payment Card Industry Data Security Standard; mandatory international security requirements for cardholder data handling.
RBAC	

Acronym / Term	Engineering Definition
	Role-Based Access Control restricting system endpoints based on defined identity classes (Customer, Merchant, Rider, Admin).
WSS	WebSocket Secure protocol establishing persistent, full-duplex bi-directional communication channels over TLS.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The platform operates as a multi-tier, highly available distributed architecture. Client layers interface through high-throughput API Gateways performing Layer-7 load balancing, SSL termination, and rate-limiting. Application logic is partitioned into microservices communicating internally via asynchronous event queues (Kafka/RabbitMQ) and synchronous gRPC/REST APIs.



2.2 User Classes and Operating Personas

- **Consumer / Customer:** Explores restaurants via spatial radius filtering, builds multi-item carts, initiates payment transactions, monitors driver GPS telemetry in real-time, and rates service. Requires high visual responsiveness (<300ms interactions).
- **Restaurant Partner:** Manages store operational hours, updates menu item availability, receives audiovisual order alerts, updates kitchen preparation milestones, and tracks business settlement summaries.
- **Delivery Fleet Partner:** Receives localized dispatch requests, reviews compensation structures, executes turn-by-turn navigation, provides real-time location telemetry, and completes drop-off verification using OTPs.
- **Platform Administrator:** Conducts KYC verification for partner restaurants and riders, audits fraud metrics, configures dynamic pricing coefficients, and handles escalated dispute workflows.

2.3 Operating Environment & Technical Stack

- **Mobile Platforms:** Android 9.0+ (API Level 28+) and iOS 15.0+ running cross-platform or native SDK runtimes.

- **Web Consoles:** Standards-compliant web browsers (Chromium 100+, Safari 15+, Firefox 100+) supporting WebSockets and Canvas/WebGL.
- **Backend Execution Environment:** Cloud-native Linux containers (Kubernetes clusters, Ubuntu 22.04 LTS baseline).
- **Persistent Storage:** PostgreSQL 15+ with PostGIS extensions for spatial data; Redis 7.x clusters for session storage, caching, and geospatial pub/sub message brokers.

3. FUNCTIONAL REQUIREMENTS

3.1 Authentication & Profile Architecture

- **FR-01 (Multi-Factor Mobile Auth):** The system shall support customer, driver, and merchant authentication via E.164 formatted mobile numbers using a cryptographically randomized 6-digit OTP expiring in 180 seconds.
- **FR-02 (Stateless Token Management):** Upon successful authentication, the system shall issue an asymmetric signed JWT (15-minute validity) alongside an encrypted HTTP-Only refresh token with automatic rotation upon reuse.
- **FR-03 (Geolocated Address Persistence):** The system shall enable customers to store distinct delivery coordinates tagged with attributes (Home , Work , Other), containing explicit building, landmark, and access notes.

3.2 Discovery, Catalog & Dynamic Inventory

- **FR-04 (Geospatial Merchant Radius Search):** The system shall query active restaurants within a bounding geodesic circle (default: 7.0 km radius) of the user's GPS position utilizing PostGIS spatial indexing.
- **FR-05 (Faceted Discovery & Search):** The system shall provide full-text and faceted search across menu items, cuisines, dietary flags (Veg, Non-Veg, Vegan), ratings (>4.0), and delivery speed within 300 milliseconds.
- **FR-06 (Real-Time Stock Availability Toggle):** The system shall allow restaurant operators to flag items or categories as Out of Stock , propagating updates to active browsing clients via Redis Pub/Sub within 2 seconds.

3.3 Cart Management, Dynamic Pricing & Billing

- **FR-07 (Single-Merchant Cart Enforcement):** The system shall enforce single-vendor cart isolation. Adding an item from a different vendor shall prompt explicit user confirmation to flush or maintain the active cart.
- **FR-08 (Algorithmic Fee Aggregation):** The checkout engine shall compute the final order total according to:

$$Total = \Sigma(Item\ Price \times Qty) + Fee_{platform} + Fee_{packaging} + Fee_{delivery}(dist, weather) + GST - Discount$$

- **FR-09 (Promotional Coupon Execution):** The system shall validate promotional coupon rules (minimum cart value, customer tier eligibility, and expiration windows) before decrementing payable balances.

3.4 Payment Processing & Financial Safeguards

- **FR-10 (Payment Gateway Orchestration):** The system shall communicate with PCI-compliant payment gateways using signed webhooks to verify transactions prior to setting the order status to Payment Confirmed .
- **FR-11 (Risk-Engine Controlled COD):** The system shall support Cash on Delivery (COD) only when user account fraud risk scores fall below threshold limits and the order amount is under £/₹ 2,000.
- **FR-12 (Automated Reversal Transactions):** In the event of restaurant rejection or delivery failure, the system shall trigger an automated payment reversal API call within 60 seconds.

3.5 Order Lifecycle & Courier Dispatch Engine

- **FR-13 (Kitchen Audio-Visual Dispatch Loop):** The restaurant console shall sound a continuous alert upon receipt of an order, enforcing an acceptance or rejection decision within a 120-second timeout window.
- **FR-14 (Heuristic Courier Allocation):** The system shall allocate orders to the optimal active delivery partner using a weighted metric:

$$Score = w_1(Distance_{rider-to-restaurant}) + w_2(Rider\ Rating) + w_3(Current\ Queue\ Depth)$$

- **FR-15 (Strict State Machine Invariance):** The order engine shall enforce unidirectional status progression:
PLACED → ACCEPTED → PREPARING → RIDER_ASSIGNED → OUT_FOR_DELIVERY → DELIVERED

3.6 Real-Time Telemetry & Tracking

- **FR-16 (High-Frequency Telemetry Ingestion):** Active delivery partner clients shall broadcast geographic telemetry (lat, lon, bearing, speed) every 5 seconds over an authenticated WSS connection.
- **FR-17 (Client Map Interpolation & Dynamic ETA):** The customer map visualizer shall render smooth vehicle marker interpolation while dynamically recalculating ETA based on routing APIs.
- **FR-18 (Drop-Off Verification):** Delivery completion requires the driver to input a 4-digit OTP provided by the customer at the time of physical handover.

4. REQUIREMENTS TRACEABILITY MATRIX (RTM)

Req ID	Requirement Description	Architectural Service	Verification Method	Priority
FR-01	Multi-Factor Mobile OTP Auth	Identity / Auth Service	Automated Unit & SMS Test	CRITICAL
FR-02	Stateless Token Management	Auth & API Gateway	Security Token Expiry Test	HIGH
FR-04	Geospatial Merchant Radius Search	Catalog / PostGIS Engine	Spatial Query Benchmark	CRITICAL
FR-06	Real-Time Inventory Toggle	Catalog / Redis Cluster	Latency & Propagation Test	HIGH
FR-07	Single-Merchant Cart Isolation	Cart & Session Service	Boundary Assertion Suite	HIGH
FR-08	Algorithmic Fee Calculation	Pricing Engine	Deterministic Ledger Tests	CRITICAL
FR-10	Payment Webhook Verification	Payment Orchestrator	Mock Payment Gateway Test	CRITICAL
FR-14	Courier Dispatch Allocation	Logistics Engine	Heuristic Dispatch Sim	CRITICAL
FR-15	Strict State Machine Invariance	Order Engine	State Transition Fuzzing	CRITICAL
FR-16	5-Sec Telemetry WSS Stream	Telemetry Ingestion	Concurrent Socket Load Test	HIGH
FR-18	Drop-Off OTP Validation	Fulfillment Service	End-to-End Verification	HIGH

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance & Throughput Standards

- **P95 Latency Thresholds:** Geospatial discovery and catalog browsing queries shall complete within ≤ 300 ms under a concurrent load of 25,000 active requests.
- **Checkout Concurrency:** The order execution engine shall support a sustained throughput of 2,000 checkout transactions per second without lock contention or deadlocks.
- **Telemetry Propagation:** End-to-end telemetry propagation delay from driver device to the observing customer map shall not exceed 1,500 ms over standard LTE/5G.

5.2 Information Security & Compliance

- **Data Transport & Ingestion:** All external communication must enforce TLS 1.3 encryption. Internal inter-service communication within Kubernetes pods must mandate mutual TLS (mTLS).

- **PCI-DSS Non-Storage Compliance:** Raw debit/credit card numbers and CVV codes shall never cross or touch platform compute instances; tokens are handled strictly by certified third-party SDKs.
- **Rate Limiting & Threat Shielding:** The edge gateway shall restrict traffic to 120 requests/minute per authenticated client ID, shielding endpoints against brute-force and DDoS vectors.

5.3 Reliability, Availability & Recovery

- **High Availability Target:** The system shall guarantee an operational SLA of 99.95% annual uptime, excluding scheduled off-peak maintenance.
- **Disaster Recovery Metrics:** Databases must support streaming cross-region replication with a Recovery Point Objective (RPO) ≤ 60 seconds and Recovery Time Objective (RTO) ≤ 5 minutes.

6. DETAILED SYSTEM USE CASES

Use Case UC-01: End-to-End Order Placement & Gateway Settlement

Primary Actors: Consumer Client, Payment Gateway API, Order Engine

Preconditions: Customer has items from a single verified merchant in cart; delivery address coordinates are locked.

Main Execution Flow:

1. Customer navigates to checkout screen; Order Engine calculates real-time delivery distance fee and applicable GST.
2. Customer selects online payment method (UPI / Credit Card) and taps "Place Order".
3. System reserves order in state `PENDING_PAYMENT` and requests a signed payment session from Gateway.
4. Customer fulfills authorization inside payment interface; Gateway returns signed webhook confirmation.
5. System validates cryptographic signature, updates state to `PAYMENT_CONFIRMED` , and triggers restaurant kitchen loop.

Exception Path (Payment Failure): If payment gateway returns timeout or authorization decline, the order reverts to `PAYMENT_FAILED` , items remain in cart, and user is presented with alternative payment options.

Use Case UC-02: Kitchen Order Processing & Handover

Primary Actors: Restaurant Operator, Kitchen Terminal System

Preconditions: Order state is `PAYMENT_CONFIRMED` .

Main Execution Flow:

1. Merchant device triggers auditory alarm and displays itemized preparation manifest.
2. Operator verifies kitchen bandwidth and taps "Accept Order" within 120 seconds.
3. Order transitions to `ACCEPTED` then `PREPARING` ; dispatch engine initiates courier proximity lookup.
4. Kitchen finishes food packaging and operator taps "Mark Ready". State advances to `FOOD_PREPARED` .

Exception Path (Timeout Rejection): If no acknowledgment occurs within 120 seconds, system automatically cancels order, records merchant penalty flag, and initiates an immediate customer refund via the payment gateway.

Use Case UC-03: Real-Time Courier Dispatch, Telemetry & Handover

Primary Actors: Delivery Partner, Dispatch Engine, Customer Client

Preconditions: Order is in `PREPARING` state; driver is active within 3 km.

Main Execution Flow:

1. Dispatch engine assigns order to optimal driver; driver receives route overview and payout quote, accepting within 30 seconds.
2. State updates to `RIDER_ASSIGNED`. Driver arrives at restaurant, confirms pickup, and updates to `OUT_FOR_DELIVERY`.
3. Driver mobile app streams GPS telemetry every 5 seconds over WSS; Customer interface displays live vehicle positioning.
4. Driver arrives at drop location; customer provides 4-digit OTP. Driver enters OTP; system confirms handover and flags order `DELIVERED`.

Exception Path (Driver Reject): If driver rejects or fails to respond within 30 seconds, dispatch cascades to next nearest active courier node.

7. DATABASE SCHEMA & UML STRUCTURAL MODELING

The platform persistence tier utilizes PostgreSQL with PostGIS extensions alongside an object-oriented domain model to manage operational consistency, entity states, and transactional lifecycles.

7.1 Object-Oriented Domain UML Class Diagram

The structural domain model below illustrates the entity classes, attributes, operations, and relationships (inheritance, associations, aggregations, enumerations) governing the Zomato system architecture:

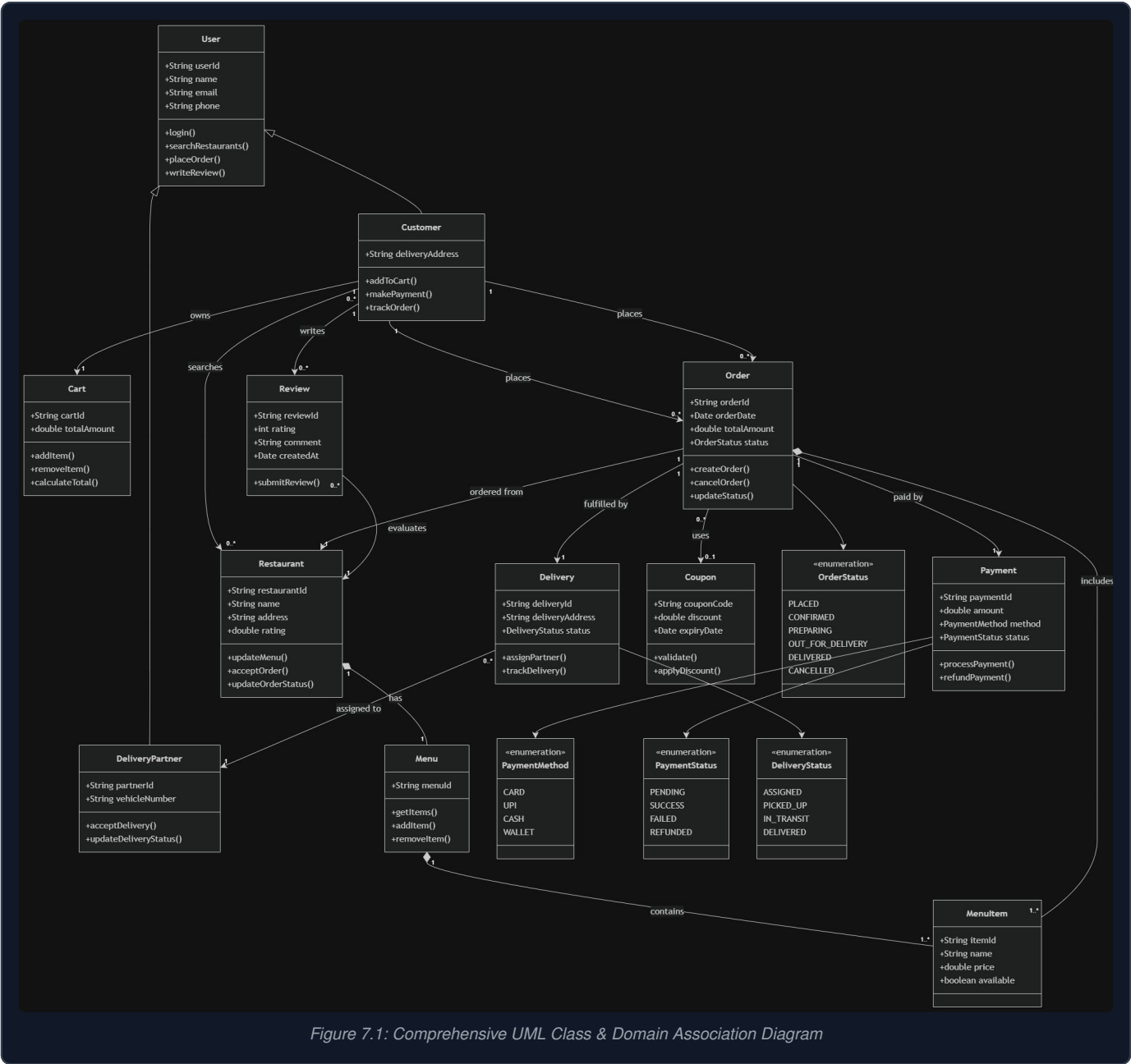


Figure 7.1: Comprehensive UML Class & Domain Association Diagram

7.2 Relational Data Dictionary & Schema Definitions

```

+-----+
|                                     |
|               users                 |
|-----+
| id           : UUID [PK]           |
| phone_number : VARCHAR(15) [UNIQUE, NOT NULL] |
| email        : VARCHAR(255) [UNIQUE] |
| role         : ENUM('customer','merchant','driver','admin') |
| is_active    : BOOLEAN [DEFAULT TRUE] |
| created_at   : TIMESTAMPTZ [NOT NULL] |
+-----+
|                                     |
|               | 1                   |
|               |                     |
|               | M                   |
+-----+
|                                     |
|               addresses             |
|-----+
| id           : UUID [PK]           |
| user_id      : UUID [FK -> users.id] |
| address_line : TEXT [NOT NULL]     |
| location     : GEOGRAPHY(Point, 4326) [NOT NULL] |
| tag          : ENUM('home', 'work', 'other') |
+-----+
|                                     |
|               restaurants            |
|-----+
| id           : UUID [PK]           |
| owner_id     : UUID [FK -> users.id] |
| name         : VARCHAR(255) [NOT NULL] |
| location     : GEOGRAPHY(Point, 4326) [NOT NULL] |
| is_open      : BOOLEAN [DEFAULT TRUE] |
| rating_avg   : NUMERIC(2, 1) [DEFAULT 0.0] |
| fssai_license : VARCHAR(50) [UNIQUE, NOT NULL] |
+-----+
|                                     |
|               | 1                   |
|               |                     |
|               | M                   |
+-----+
|                                     |
|               menu_items            |
|-----+
| id           : UUID [PK]           |
| restaurant_id : UUID [FK -> restaurants.id] |
| name         : VARCHAR(255) [NOT NULL] |
| base_price    : NUMERIC(10, 2) [NOT NULL] |
| is_veg       : BOOLEAN [NOT NULL] |
| is_available : BOOLEAN [DEFAULT TRUE] |
+-----+
|                                     |
|               orders                |
|-----+
| id           : UUID [PK]           |
| customer_id  : UUID [FK -> users.id] |
| restaurant_id : UUID [FK -> restaurants.id] |
| driver_id    : UUID [FK -> users.id, NULLABLE] |
| status       : ENUM('placed','accepted','preparing', |
|                   'rider_assigned','out_for_delivery', |
|                   'delivered','cancelled') |
| total_amount : NUMERIC(10, 2) [NOT NULL] |
| delivery_fee : NUMERIC(10, 2) [NOT NULL] |
| drop_location : GEOGRAPHY(Point, 4326) [NOT NULL] |
| delivery_otp  : VARCHAR(4) [NOT NULL] |
| created_at   : TIMESTAMPTZ [NOT NULL] |
+-----+
|                                     |
|               | 1                   |
|               |                     |
|               | M                   |
+-----+
|                                     |
|               order_items           |
|-----+
| id           : UUID [PK]           |
| order_id     : UUID [FK -> orders.id] |

```

menu_item_id : UUID [FK -> menu_items.id]	
quantity : INT [NOT NULL]	
price_per_unit : NUMERIC(10, 2) [NOT NULL]	
+-----+	
+-----+	
payments	
+-----+	
id : UUID [PK]	
order_id : UUID [FK -> orders.id]	
transaction_ref: VARCHAR(255) [UNIQUE, NOT NULL]	
payment_method : ENUM('card','upi','wallet','netbanking','cod')	
payment_status : ENUM('initiated','success','failed','refunded')	
amount : NUMERIC(10, 2) [NOT NULL]	
updated_at : TIMESTAMPTZ [NOT NULL]	
+-----+	

8. VERIFICATION & ACCEPTANCE CRITERIA

Test Target	Test Condition	Expected Acceptance Benchmark
Geospatial Spatial Query	Customer invokes search query at coordinates (12.9716, 77.5946).	Query returns all open merchants within geodesic radius ≤ 7.0 km; execution latency ≤ 250 ms.
Concurrency & Cart Race	100 concurrent requests compete for 1 unit of available inventory.	Exactly 1 order completes successfully; 99 requests fail with HTTP 409 Conflict without data corruption.
Telemetry Streaming Load	5,000 active delivery drivers streaming coordinate packets every 5 seconds.	Zero connection socket drops; P99 delivery of map coordinate updates to subscribers ≤ 450 ms.
Order State Machine Invariant	Attempt illegal transition from PLACED directly to DELIVERED .	API returns HTTP 400 Bad Request; order state remains unchanged; audit log records security anomaly.

9. STANDARDS AND REFERENCES

- 1. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications, IEEE Computer Society.
- 2. ISO/IEC/IEEE 29148:2018: Systems and software engineering — Life cycle processes — Requirements engineering.
- 3. PCI Security Standards Council: Payment Card Industry Data Security Standard (PCI-DSS) v4.0, 2022.
- 4. RFC 6455: The WebSocket Protocol, Internet Engineering Task Force (IETF).
- 5. OpenGIS Consortium: Spatial Reference System Specifications (EPSG 4326 - WGS 84).